Version 1

Jigawa State Election Situation Room — Comprehensive & Articulated Framework

Below is a complete, non-technical, implementation-ready framework for a three-layered **Jigawa State Election Situation Room** (web + Android + iOS + SMS). It focuses on roles, visibility, data flows, UX rules, governance, operations, security, reporting, rollout and support — all described as policies, processes and deliverables (no code).

1. Purpose & scope

Purpose: Provide a single, auditable, real-time system to collect, monitor, analyse and respond to ward-level election information across Jigawa State (27 LGAs, 287 wards).
Scope: Field reporting (ward agents), supervisory monitoring, state-level oversight, real-time alerts, analytics dashboards, SMS gateway, print/export reports and incident escalation.

2. Three-Layer Access & Responsibilities (Logging / Visibility model)

Layer 1 — Super Admin (State Situation Room)

Primary responsibilities

- Full system control: create/disable accounts, set roles, manage master data (LGAs, Wards, Parties, Senatorial Districts, Candidates).
- Configure system settings: reporting windows, correction window, SMS templates, push notification channels, access control rules.
- View and export all data, run state-level analytics, generate official reports and signed printouts.
- Review audit logs and perform compliance/forensic review.

Visibility & permissions

- Full visibility across all LGAs/wards, all reports, all audit logs, user activities and SMS flows.
- Create or update users for any role.

Logging

- All actions recorded with who, when, where (LGA/Ward context if relevant), action detail and device/ip metadata.

Layer 2 — Supervisor (Call Desk / LGA Coordinators)

Primary responsibilities

- Monitor incoming reports and messages from assigned LGAs or statewide (depending on assignment).
- Validate, flag, escalate incidents and annotate or request clarifications from Ward Agents.
- Send SMS replies or push notifications to ward agents and other stakeholders.
- Curate and prepare consolidated reports for Layer 1.

Visibility & permissions

- Read access to all reports inside assigned scope (LGA or state-level if designated).
- Can view audit logs only for their scope.
- Can send SMS (scoped) and perform limited edits (append notes, mark verified or escalated).

Logging

- Supervisor actions logged with context and justification (required when editing or escalating).

Layer 3 — Ward Agent (Field Reporter)

Primary responsibilities

- Submit ward-level reports via mobile app, web form or SMS using the structured format. Attach media as required.
- Respond to supervisor queries and confirm corrections during the correction window.
- Provide authentication (phone + app login) and basic agent meta-data.

Visibility & permissions

- See own submissions and system messages/alerts only.
- Cannot view other agent's personally-identifiable activity or other wards' reports.

Logging

- Every submission is logged automatically with timestamp, device id, and submission method (web/app/sms).

3. Core Data & Reporting Structure (described)

Master reference data

- **27 LGAs** — each LGA has an ID, name, supervisor(s) and contact list.
- **287 Wards** — each ward record maps to an LGA and a senatorial district.

- **Political Parties** — party master list and candidate master list (for each post: senatorial, federal, state, local).
- **Users/Agents** — profile with role, gender, phone, LGA, Ward, party affiliation (if applicable) and active status.
- **Security types** — canonical list: Police, Civil Defence, NSCDC, Army, Vigilante, Other.

Report record (fields to capture each submission)

- **Meta (auto)**: Agent name, agent ID, LGA, Ward, timestamp (auto).
- **Part B — Agents & Security**: Party registered agent (name/id), INEC agent name, counted voter papers (number), security present? (Y/N), security type(s).
- **Part C — Voting counts**: Time voting closed (if available),
    - number who cast votes,
    - voting papers left (not cast),
    - valid ballot paper count,
    - invalid ballot paper count.
- **Optional**: Media attachments
    - (photos/videos),
    - free-text remarks,
    - submission method (web/app/sms),
    - sync status (for offline uploads).
- **Administrative fields**: Report status;
    - draft,
    - submitted,
    - verified,
    - flagged,
    - escalated),
    - audit trail entries (who edited, when, what).

Data validation rules (process, not code)

- Required: - agent identity – LGA - Ward - submission timestamp.
- Numeric fields: must be non-negative integers.
- Consistency check: valid + invalid should be <= number cast votes. Flag and hold for supervisor review if inconsistent.
- If security present = Yes then at least one security type must be provided.

4. Data capture workflows

Primary workflow (ward agent)

1. Open app/web → authenticate → form opens prefilled with agent profile (name, LGA, Ward).
2. Fill Part B (party agent, INEC agent, counted papers, security).
3. Fill Part C (voting closed time, counts) and attach evidence.
4. Auto timestamp recorded. Validate locally; show warnings if counts inconsistent.
5. Submit: system assigns status submitted. Agent receives acknowledgment (in-app + SMS).
6. If offline: Save as draft; app auto-syncs when network returns and logs sync_time.

Supervisor workflow

1. Inbox shows new reports sorted by severity/time.
2. Quick actions: verify, flag (requires reason), escalate (triggers SMS/email to predetermined list), request clarification (sends question to agent).
3. Supervisor can add annotations that are appended to audit log.
4. After verification or escalation, the status is updated and reflected to Layer 1.

Super Admin workflow

1. Monitor state dashboards and escalations.
2. Adjust configuration (e.g., change correction window).
3. Approve deletions (if policy allows) and run official export/print.

5. Dashboards & Analytics (what each should show)

Global filters (always available)

- Date/time range, Submission method (web/app/sms), LGA, Ward, Party, Candidate, Gender, Logged In status (Logged in / Not logged in), Security present (Y/N), Report status.

Dashboard panels (key)

1. **Agent Login Status** — drop-down filter: Logged in / Not logged in; count of agents per LGA and ward, with list of non-logged-in agents (for outreach).
2. **Submissions by Ward** — table with number of reports, latest report time; ability to drill to each report.
3. **By Gender** — count of agents and reports submitted by Male / Female / Other.
4. **By Local Government** — aggregated counts and incident heatmap by LGA.
5. **By Party** — stacked totals of votes and reports by party.
6. **By Senatorial District** — roll-up summary and notable flags.

7. **Candidates Distribution** — per-candidate reported totals across wards, breakdown by party (for verification).
8. **Security Incidents** — list & map of reports with security_present = true and their security types.
9. **Media Gallery** — latest media attachments for validation (with verification flags).
10. **Real-time Feed / Inbox** — incoming reports and SMS queue with quick actions.

Print & Export options

- Export filters into CSV or PDF with header metadata (selected filters, generated by, timestamp).
- Print-ready reports: agents by ward, by gender, by LGA, by party, by senatorial district, candidate distribution.

6. SMS Integration & Operation (policy & templates)

Goals

- Allow agents to submit minimal reports via SMS and receive confirmations.
- Allow supervisors & super admin to send group SMS announcements and targeted replies.

Allowed SMS operations

- **Inbound**: SMS short-format report (predefined syntax) OR SMS ID to request full form link.
- **Outbound**: Confirmation messages, escalation alerts, broadcast advisories to defined groups.

SMS submission format (policy)

- Use a concise structured template easy to memorize. Example structure (communicated in training):
    - JIG REPORT <WardCode> <AgentInitials> <PartyCode> <CountedPapers> <Valid> <Invalid> <Sec:Y/N> <SecCodes>
- Example incoming SMS: JIG REPORT W087 MA APC 120 118 2 Y POLICE, VIG
- On receipt: system parses, creates a report with submitted_via = SMS, timestamps it and replies with CONF IR <ReportID> Received. Thank you.

SMS security & confirmations

- Only registered phone numbers mapped to agent accounts accepted as authenticated SMS senders.

- SMS parsing failures trigger an automated reply asking user to resend in correct format or use mobile app.

7. Offline-first & synchronization policy (field operations)

Offline capture

- Mobile apps must allow drafting forms offline with local store.
- All drafts carry device metadata and local timestamp.

Sync rules

- On reconnect, app submits batch to server; server assigns authoritative timestamp for reporting but keeps agent's local timestamp as submitted_at_device.
- If server detects conflicting edits (two different submissions for same report id), flag for supervisor review.
- Sync success/failure and last sync time shown to agent in UI.

Data integrity

- Record submitted_via and device_id for every record to support verification.
- Media uploads may be queued and referenced with placeholders until upload completes; supervisor can view "pending media" tag.

8. Audit & Accountability (mandatory policy)

- **Immutable audit trail** for every create/update/delete action containing: actor, role, timestamp, IP/device info, previous value and new value, and reason for change (for edits).
- **Edit windows**: configure correction window (e.g., 10–30 minutes) where agent may amend report — all edits retained in audit log and visible to supervisors and admins.
- **Approval trail**: Supervisor verification must be logged — Verification record: who verified, when, notes.
- **Regular audits**: Scheduled audits of randomly sampled reports to check for anomalies and training needs.

9. Security & Privacy (policy-level)

Authentication & access control

- Multi-factor auth for Super Admin and optional for Supervisors.

- Strong password rules and forced periodic password changes.
- Role-based access with row-level visibility restrictions (Ward Agents only their own ward, Supervisors to assigned LGAs).

Data protection

- Personal identifiable information (PII) limited to agent contact info and necessary identifiers.
- Media and sensitive content stored securely and accessible only to authorized roles.
- Data retention policy defined (e.g., archive after X years) and legal compliance for election data handling.

Operational security steps

- Enforce HTTPS across web and API endpoints.
- Use device binding for agent app (phone number + device identifier).
- Rate-limiting and anomaly detection for bulk submissions or suspicious patterns.

10. Governance, SOPs & Roles outside the system

Steering committee (recommended)

- Composition: State Chairman (Chair), Representatives from INEC liaison, Security Coordinator, Civil society representative, Tech Assistants, Communications lead.
- Roles: Approve SOPs, escalation ladders, data retention policy and public communications.

Standard Operating Procedures (SOPs) to write

- Agent enrolment & deactivation process.
- Report submission & correction policy.
- Escalation ladder for security incidents (who to notify, timelines).
- Media handling & evidence preservation.
- Communications & public release policy for official summaries.

Escalation matrix (example)

- **Level 1 (Supervisor)** — within 10 minutes of incident flag: call agent, confirm facts, send SMS to LGA coordinator.
- **Level 2 (LGA Coordinator)** — within 30 minutes: coordinate with local security actors, prepare consolidated report.
- **Level 3 (State/Incident Response)** — immediate for violence/major incidents: SMS + Email + Phone to State Security and Steering committee; prepare public advisory.

11. Training, On-boarding & Change Management

Training program

- **Role-based training modules** for Super Admin, Supervisors, Ward Agents.
- Training content: account setup, form filling, media capture guidelines, SMS format, offline sync, escalation, basic troubleshooting.
- Deliverables: Quick-reference card (SMS formats & codes), step-by-step screenshots, short video walkthroughs.

Pilot & phased roll-out

- Pilot in 2 LGAs (choose one urban, one rural) for user acceptance testing and to refine offline behavior and SMS parsing.
- Phase 2: Rollout remaining 25 LGAs in waves (5–7 LGAs per wave).
- Final phase: full state rollout + public communications.

Support model

- 24/7 on-call tech support during election days (rotating shifts).
- Local LGA technical champions for first-level support.
- Knowledge base (FAQs) and in-app help.

12. Reporting & KPI (what to measure)

Operational KPIs

- Technical working group.
- % of Ward Agents who successfully logged in at least once before election day.
- Average time from submission to Supervisor acknowledgment.
- % of reports containing media attachments.
- Number of escalations and response times by level.
- SMS delivery success rate and parsing success rate.

Data quality KPIs

- % of submissions failing validation rules (e.g., invalid sums).
- Number of edits per report (indicates training needs or ambiguity).
- Discrepancies found during audits (percentage).

13. Evidence & Media handling policy

- Minimum requirements for media (photo resolution, timestamp visible if possible).

- When images/videos contain violent or sensitive content, mark as sensitive and restrict viewing to authorized roles only.
- Store original media and generate thumbnails for quick review.
- Chain-of-custody notes added for media when used as evidence.

14. Print & Official Reports (policy & structure)

- **Official report template** includes: header (Situation Room name), filters used, time window, generator name, signature block, summary metrics, key incidents and attachments reference.
- Provide customizable print queries for:
    - Agents list by ward
    - Submissions by Gender
    - Submissions by Ward/LGA
    - Submissions by Party
    - Senatorial district consolidations
    - Candidate distribution across wards

15. Monitoring, Logging & Post-election Review

Monitoring

- Monitor system health: uptime, API latency, SMS queue backlog, DB availability.
- Real-time alerts for system failures to tech support and Super Admin.

Post-election review

- Prepare a data handover pack: raw reports, audit logs, media, exports and summary analytics.
- Conduct a lessons-learned workshop with logs and KPIs, producing an after-action report with improvement plan.

16. Implementation governance & immediate deliverables (what to prepare now)

Documents to produce before development:

1. Role & permission matrix document (finalized).
2. Detailed data dictionary listing every report field and allowed values.
3. SMS format guide and test cases.
4. SOPs for enrollment, escalation, audit and media handling.
5. Training curriculum and quick-reference cards.

6. Pilot plan with chosen LGAs, acceptance criteria and evaluation metrics.
7. User acceptance test (UAT) checklist and sample test cases.
8. Communication plan for stakeholders (INEC liaison, security, civil society).

Templates to prepare (non-technical):

- Agent enrollment form.
- Incident escalation form.
- Official PDF report template.
- Media consent and privacy notice (for how photos will be used).

17. Risks & mitigation (top items)

1. **Poor network coverage in rural wards** — Mitigation: offline-first app with robust sync and SMS fallback.
2. **Agent data entry errors** — Mitigation: form validation, training, audit checks and short correction window.
3. **SMS parsing failures** — Mitigation: clear SMS templates, auto-reply with correction guidance, parser tolerant to common errors.
4. **Unauthorized data access** — Mitigation: role-based access controls, device binding, MFA for high-privilege users, immutable audit logs.
5. **False or intentionally manipulated reports** — Mitigation: media evidence requirement, random audit sampling, cross-verification by supervisors.

18. Operational checklist for election day (compact)

- ☐ All 287 Ward Agents enrolled, verified and tested (login + sample submission).
- ☐ Supervisors assigned to each LGA with contact details.
- ☐ SMS gateway tested (send/receive) with sample messages.
- ☐ Mobile app installed & offline sync tested on representative devices.
- ☐ Super Admin credentials secured and MFA enabled.
- ☐ Escalation contact list published and tested.
- ☐ Backup/restore and monitoring configured.
- ☐ Tech support rota scheduled.

19. Next practical outputs I can produce immediately (you pick one)

(They will be delivered as non-code documents or designs.)

- A one-page **Operational SOP** for ward agents (step-by-step instructions + SMS cheat sheet).
- A **Dashboard content spec** (visual description of each widget and its filters) for the designer.
- A **Training slide deck outline** for a 90-minute Ward Agent training.
- A **Pilot plan** for rolling out in 2 LGAs (objectives, acceptance criteria, timeline, test cases).
- A **Printable report template** (structured fields to copy into your document editor).

20) Role & Permission matrix (visibility & actions)

Role / Layer	Login	Create accounts	View all reports	View inbox/messages	Submit ward reports	Edit reports	Run/Print reports	SMS send
Super Admin (Layer 1)	✓	✓	✓	✓	✓ (any)	✓	✓	✓
Supervisor (Layer 2)	✓	✗	✓ (read-only; limited edits)	✓ (full)	✓ (if assigned)	Limited (flag/annotate)	✓	✓ (scoped)
Ward Agent (Layer 3)	✓	✗	Own submissions only	✓ (system alerts only)	✓	✗ (unless correction within time window)	✗	View only / opt-in

Visibility rules

- Layer 3 sees only their own submissions + system messages.
- Layer 2 sees all submissions across LGAs they are assigned to (or entire state if Supervisor).
- Layer 1 sees everything and system logs, user accounts, settings.

Audit log layers

- **Layered logs** saved for every action: created_by_role, created_by_user_id, timestamp, ip, device, previous_value (for edits). Super Admin can view all logs; Supervisor can view logs for LGAs assigned; Ward Agent can only see audit of their own entries.

21) Data model (tables + fields) — core entities

I'll give key tables and sample SQL CREATE TABLE snippets you can copy into Postgres / MySQL.

22) Data capture UI — form structure & validations

A single multi-part form with auto-save and attachments. Show required/optional.

Part A — Identity & meta (auto)

1. Name of Agent — prefilled from account (editable if allowed)
2. Local Government — prefilled from account (dropdown if admin)
3. Ward — prefilled from account
4. Time of Reporting — auto timestamp, shows user timezone; allow manual edit only for corrections with audit.

Validation: agent_name required, lga_id & ward_id required.

Part B — Agents & security

1. Select Name of Party Registered Agent — dropdown of users assigned as party agents (nullable)
2. Name of INEC Agent — free text (optional)
3. Number of Counted Voters Paper — integer >= 0
4. Are there security Personnel? — dropdown Yes/No (required)
5. If Yes, Security Type — multi-select: Police, Civil Defense, Vigilante, NSCDC, Army, Other (store as JSON)

Validation: If security_present = Yes, require at least one security type.

Part C — Voting counts

1. Time Voting Closed — time picker (or null)
2. Number of persons that cast votes — integer >= 0
3. Voting Papers Left Not Cast — integer >= 0
4. Valid Ballot Paper — integer >= 0
5. Invalid Ballot Paper — integer >= 0

Validation: Sum(valid + invalid) <= number_cast_votes (warn if inconsistent). Optionally require counted_voter_papers >= number_cast_votes.

Part D — Media & remarks + submit

- Attach photos (min 0, max 5), video (optional), audio (optional).

- Remarks (free text)
- Save draft / Submit button (once submitted, allow correction window configurable — e.g., 10 minutes).

SMS submission

- Support SMS shortcodes for minimal reports (see SMS section below). SMS submissions map to the same fields; parse and create ward_reports entries; send SMS confirmation back.

23) Dashboards & analytics (filters & widgets)

Dashboards must be filterable by date/time ranges and have the following filter facets: Logged In Status, Gender, Ward, LGA, Party, Senatorial District, Party Candidates.

Primary widgets

1. **Agents Login Status**: dropdown filter (Logged in / Not logged in) + counts per LGA/ward.
2. **Agents by Gender**: donut chart (Male / Female / Other).
3. **Submissions by Ward**: sortable table (ew: count, latest report time).
4. **Submissions by LGA**: choropleth or list.
5. **By Party**: stacked bar showing votes / reports per party.
6. **By Senatorial District**: aggregated key stats.
7. **Candidate Distribution**: per-party candidate vote totals (sortable).

Search & drilldown

- Global search for agent name, phone, ward, report id.
- Click into any widget to drill down (e.g., click LGA -> list of wards -> list of reports -> individual report view).

Special dashboards for Supervisors

- **Inbox Page**: chronological queue of incoming reports & SMS with quick actions (flag, escalate, call, send SMS reply).
- **Security incidents**: filter by security_present = true and show heatmap by ward.

Data export & print

- Report printing by query as you requested: queries to produce lists/filters for (agents by wards, gender, wards, local gov, parties, senatorial, candidates).
- Allow exports: CSV for spreadsheets; PDF for printable reports with header (Election Situation Room, Jigawa State), filter details, and sign-off area.

24) Key API endpoints (examples)

RESTful endpoints (or GraphQL if preferred). Example JSON snippet and endpoints (authentication via JWT / OAuth2).

Authentication / users

- POST /api/auth/login → returns JWT
- POST /api/users (super-admin)
- GET /api/users?role=&lga=&ward= — list users

Reports

- POST /api/reports — create ward report (body = Part A/B/C)
- GET /api/reports/:id
- GET /api/reports?lga=&ward=&party=&from=&to=&security_present=&agent_id= — filtered reports
- PUT /api/reports/:id — edit (with audit)

Dashboard

- GET /api/dashboard/summary?from=&to=&lga=&ward=
- GET /api/dashboard/agents/login-status

SMS

- POST /api/sms/send — send SMS
- POST /api/sms/receive — webhook from SMS provider to ingest incoming SMS

Audit

- GET /api/audit?entity=&user_id=&from=&to=

25) SMS integration (send + receive)

Support both sending SMS (outbound) and receiving inbound SMS mapped to reports.

Recommended flow

- Use providers with Nigerian coverage: e.g., Africa's Talking, Twilio, MessageBird — integrate via provider SDK/webhooks.
- **Outbound SMS**: POST /api/sms/send (body: recipients[], message, sender_id, template_id). Super Admin + Supervisor can send; Ward Agents receive confirmations.
- **Inbound SMS**: provider calls POST /api/sms/receive with SMS text. Implement lightweight parser to map to fields. Example simple command format:

    - JIG REPORT <ward_code> <agent_name> <party_code> <counted_papers> <valid> <invalid> <security:Y/N> <security_types>
    - E.g. JIG REPORT W-012 Musa APC 120 118 2 Y POLICE,VIG
- Parse and create ward_reports with submitted_via = 'sms'. Send automatic confirmation SMS with report id.

Delivery receipts & retries

- Use provider callbacks to update status; store SMS send status & retry failed messages.

26) Offline-first & sync (critical for field agents)

- Mobile app caches forms locally using SQLite; allow draft saves.
- When network restored, app posts batched reports to POST /api/reports/batch.
- Use optimistic UI and conflict resolution: server enforces last-write-wins or requires approvals for conflicting edits (flag to Supervisor).
- Track sync_status and device_id in ward_reports.

27) Security, privacy & compliance

- Enforce HTTPS (TLS 1.2+).
- Store passwords hashed (bcrypt/argon2).
- JWT for auth with short expiration + refresh tokens.
- Role-based access control (RBAC) and row-level security (RLS) in DB for strict data access by LGA/Ward.
- Audit logs immutable (append-only); store previous_value for changes.
- Media stored in secure object storage (S3 or compatible) with signed URLs and TTL.
- Data retention policy: define how long reports are retained; provide admin tools to purge older data according to electoral law.
- Two-factor auth for Super Admin & Supervisor (SMS/Authenticator).
- Rate-limit endpoints & input sanitization to prevent injection.

28) Printing & reporting queries (deliverables you asked)

Reports should be queryable via the UI and printable:

Report types to support (Part D)

1. **List of agents by ward** — filter users where role = ward_agent & ward_id = X.
2. **By Gender** — SELECT gender, COUNT(*) FROM users GROUP BY gender.

3. **By Wards** — SELECT ward_id, COUNT(*) FROM ward_reports GROUP BY ward_id.
4. **By Local Government** — aggregated by lga_id.
5. **By Parties** — joins to parties table.
6. **By Senatorial District** — map wards to districts or provide mapping table and group by district.
7. **Candidate distribution** — join candidate table with ward_reports and aggregate totals.

Sample SQL to export agents not logged in

```
SELECT u.id, u.full_name, u.phone, l.name AS lga, w.name AS ward, u.last_login
FROM users u
LEFT JOIN lgas l ON u.lga_id = l.id
LEFT JOIN wards w ON u.ward_id = w.id
WHERE u.role_id = 3 -- ward agent
AND (u.last_login IS NULL OR u.last_login < (NOW() - INTERVAL '24 hours'));
```

Print

- Generate PDF from server-side rendering (e.g., wkhtmltopdf) or client-side (PWA).
- Include header with selected filters + generation timestamp + signature block for Super Admin.

28) Notifications & escalation flows

- **Real-time**: WebSocket notifications for Supervisors when security_present = true or certain keywords in remarks.
- **Escalation**: Supervisor marks incident -> automatic SMS to LGA coordinator + email + push notification.
- **Acknowledgement**: Ward agent receives confirmation SMS + web notification after submission.

29) UI/UX suggestions (brief)

- Clean dashboard with left nav: Dashboard, Inbox, Submit Report, Agents, Reports, Settings.
- Use large action buttons for Submit Report on mobile.
- In Inbox, each row shows: agent name, ward, timestamp, quick actions (flag, call, SMS).

- Filters on top bar: Date range, LGA, Ward, Party, Security present, Submitted via (web/sms/mobile).
- Use color-coded chips for status: New (blue), Flagged (yellow), Escalated (red), Verified (green).

30) Tech stack (recommended)

- **Frontend**: React (web), React Native / Flutter for mobile or PWA for faster deployment.
- **Backend**: Node.js (NestJS / Express) or Python (Django/DRF). Use GraphQL if many complex queries needed.
- **DB**: PostgreSQL (for JSONB + strong relational support).
- **Cache / realtime**: Redis + WebSockets (Socket.IO or Pusher)
- **Storage**: S3-compatible (AWS S3, DigitalOcean Spaces)
- **SMS**: Africa's Talking / Twilio (webhook support)
- **Auth**: OAuth2 / JWT, use Auth0 optionally
- **Maps / UI**: Mapbox or Leaflet (if mapping wards)
- **Hosting**: Dockerized, Kubernetes or managed services (Heroku, Railway, AWS ECS)

31) Implementation checklist (actionable milestones)

1. Create DB schema & seed LGAs (27) and Wards (287).
2. Implement auth & roles.
3. Build report submission API & frontend form (Parts A–D).
4. Implement SMS webhook parser & confirmation flow.
5. Build Inbox & Supervisor workflows + audit logging.
6. Create dashboards with the required filters and export/print functions.
7. Add offline mobile storage & sync.
8. Security review & pen test.
9. Train users & run pilot in 2 LGAs, then roll out.

32) Example JSON payload — creating a report


```
{
  "agent_user_id": 1234,
  "agent_name": "Musa Ahmed",
```

```
  "lga_id": 5,
  "ward_id": 87,
  "party_agent_id": 2001,
  "inec_agent_name": "Hassan Bello",
  "counted_voter_papers": 150,
  "security_present": true,
  "security_types": ["Police", "Vigilante"],
  "voting_closed_time": "2025-11-28T16:30:00+01:00",
  "number_cast_votes": 148,
  "voting_papers_left": 2,
  "valid_ballot_papers": 146,
  "invalid_ballot_papers": 2,
  "remarks": "Smooth process. Two spoilt ballots.",
  "media_attachments": ["https://storage.example.com/reports/1234/photo1.jpg"],
  "submitted_via": "android"
}
```


---

33) Governance & training

- Provide user manuals (Super Admin, Supervisor, Agent).
- Run brief training for Ward Agents focused on form fields, photo evidence, and SMS format.
- A quick FAQ inside app and escalation phone numbers.

34) Scalability & monitoring

- Add Prometheus + Grafana for server metrics.
- Log aggregator (ELK stack / Logflare) for application logs + audit.
- S3 lifecycle rules for media to control costs.

35) Final notes & deliverables you can copy/paste

- I included DB schema snippets, API endpoints, UI/UX guidance, SMS flows, security, offline sync, and sample queries for the print/export features you listed.
- If you want, I can:

- Produce a **detailed ER diagram** (PNG/SVG),
- Generate **Postgres migration scripts** for full schema,
- Create a **clickable Figma-style UI mockup** (text descriptions to hand to designers),
- Or produce a **working Postman collection** for the listed endpoints.

Tell me which of those you want next and I'll output it immediately (no waiting).

Comprehensive list of the 287 political wards (also known as Electoral Wards), (North West, North East, and North Central) in Jigawa State, Nigeria, organized by their 27 Local Government Areas (LGAs).

Auyo LGA

1. Auyakayi
2. Auyo
3. Ayama
4. Gafaya
5. Gamsarka
6. Gatafa
7. Gorgor
8. Kafur
9. Unik

Babura LGA

10. Babura
11. Burdugau
12. Dankumbo
13. Gidan Duna
14. Gwangwan
15. Jigawa
16. Kanya
17. Kuzunzumi
18. Kyambo

Biriniwa LGA

19. Biriniwa
20. Diginsa
21. Fagi
22. Gujungu
23. Machinamari
24. Matamu
25. Nguwa
26. Sarmai
27. Tumbi

Birnin Kudu LGA

28. Birnin Kudu
29. Kanti
30. Kawaya
31. Kudai
32. Lafa
33. Surko
34. Wurno
35. Yalwan Damai

Buji LGA

36. Amaryawa
37. Buji
38. Chamo
39. Gantsa
40. Gawar
41. Kukuma
42. Kor
43. Tsangarwa

Dutse LGA

44. Chamo
45. Duru
46. Dustsen Ma
47. Dustsen Uku
48. Jigawar Tsada
49. Kachi
50. Kargo
51. Madobi
52. Sakwaya

Gagarawa LGA

53. Gagarawa Gari
54. Gagarawa Tasha
55. Kore Balatu
56. Maikilili
57. Medu
58. Yalwa

Garki LGA

59. Buduru
60. Garki
61. Kanya
62. Kore
63. Muku
64. Raba
65. Sabon Garin
66. Wa

Gumel LGA

67. Baikarya
68. Dan Ama
69. Garin Gambo
70. Gumel
71. Hausawa
72. Kofar Arewa
73. Kofar Yamma
74. Sabon Garin

75. Zango

Guri LGA

76. Abunabo
77. Adiyani
78. Garbagal
79. Guri
80. Kwaila
81. Margadu
82. Matara Baba

Gwaram LGA

83. Fagam
84. Gwaram
85. Kila
86. Kwanda
87. Maruta
88. Sara
89. Surko
90. Zandam

Gwiwa LGA

91. Buntusu
92. Dabi
93. Guntai
94. Gwiwa
95. Korau
96. Ruruma
97. Yola

Hadejia LGA

98. Atafi
99. Dubantu
100. Fagi
101. Kasuwar Kofa
102. Kasuwar Kuda
103. Majema
104. Matsaro
105. Rumfa
106. Sabon Garu

Jahun LGA

107. Aujara
108. Daba
109. Gangan
110. Garki
111. Gushuari
112. Hauma
113. Jahun

114. Kanwa

Kafin Hausa LGA

115. Azaga
116. Bulunchai
117. Daba
118. Daba Mashema
119. Fagi
120. Faragwai
121. Kafin Hausa
122. Kwazalewa
123. Sara

Kaugama LGA

124. Askandu
125. Daba
126. Dagaro
127. Jarkasa
128. Kaugama
129. Karkara
130. Marke
131. Urmi

Kazaure LGA

132. Ba'auzini
133. Daba
134. Gada
135. Ganga
136. Janbiri
137. Kanti
138. Roni
139. Sabaru
140. Yandamo
141. Yankwashi

Kiri Kasamma LGA

142. Balangu
143. Fage
144. Kiri
145. Kwara
146. Sansani
147. Siyori
148. Tasha

Kiyawa LGA

149. Andaza
150. Fateka
151. Fuskam
152. Katanga

153. Kiyawa
154. Kwanda
155. Kiyawa Tasha
156. Shuwarin

Maigatari LGA

157. Balarabe
158. Danladi
159. Fulata
160. Jajire
161. Kukayasku
162. Maigatari
163. Matoya
164. Tumbi

Malam Madori LGA

165. Arki
166. Dangwaleri
167. Falu
168. G/Galadima
169. Garin Chiroma
170. Kachallari
171. Kafin Babushe
172. Kargo
173. Kukadi
174. Malam Madori
175. Togawa

Miga LGA

176. Abalago
177. Hantsu
178. Koya
179. Miga
180. Sabon Garin
181. S/Tsaunuka
182. Tsakuwa
183. Zareku

Ringim LGA

184. Chai-Chai
185. Kafin Baka
186. Karshi
187. Kyambo
188. Pale
189. Ringim
190. Sansani
191. Sintilmawa
192. Tofa

193. Yandutse

Roni LGA

194. Dansure
195. Fara
196. Gora
197. Kwaita
198. Roni
199. Sankau
200. Yanzaki
201. Zango

Sule Tankarkar LGA

202. Albasu
203. Amanga
204. Arbus
205. Danladi
206. Dantanoma
207. Sule Tankarkar

Taura LGA

208. Badage
209. Gurduba
210. Jarkasa
211. Kwalam
212. Mudantha
213. S/Safiya
214. Taura

Yankwashi LGA

215. Amaryawa
216. Dankumbo
217. Duduri
218. Gada
219. Karkarna
220. Ringim
221. Yankwashi

List of Newly Created Wards (Carved out from existing ones)

From Birnin Kudu LGA:

222. Kafin Babushi
223. Unguwar Jibrin

From Buji LGA:

224. Fajingi

From Dutse LGA:
225. Daga
226. Jigawar Sarki

From Garki LGA:
227. Tsangarwa

From Gumel LGA:
228. Danzomo
229. Galagamma

From Guri LGA:
230. Kargo

From Gwaram LGA:
231. Guda

From Gwiwa LGA:
232. Darina

From Hadejia LGA:
233. Unguwar Gabas
234. Unguwar Yamma

From Jahun LGA:
235. Gatafa

From Kafin Hausa LGA:
236. Kandawa
237. Kazura

From Kaugama LGA:
238. Galagamma

From Kazaure LGA:
239. Daba
240. Gada
241. Ganga
242. Yandamo

From Kiri Kasamma LGA:
243. Kanti

From Kiyawa LGA:
244. Daba
245. Sabon Garin

From Maigatari LGA:
246. Madana
247. Tandama

From Malam Madori LGA:
248. Danlawal
249. Kachallari
250. Kafin Babushe

From Miga LGA:
251. Kunkun
252. Sabon Garin Miga

From Ringim LGA:
253. Kafin Baka
254. Ringim
255. Sintilmawa

From Roni LGA:
256. Bagada
257. Roni
258. Yanzaki

From Sule Tankarkar LGA:
259. Sule Tankarkar

From Taura LGA:
260. Jarkasa
261. Kwalam
262. Mudunta
263. Taura

From Yankwashi LGA:
264. Amaryawa
265. Dankumbo
266. Duduri
267. Gada
268. Karkarna
269. Ringim
270. Yankwashi

Additional New Wards to reach the final count:

(These are further subdivisions or reclassifications from various LGAs)

271. Garki (from Garki LGA)
272. Kanya (from Garki LGA)
273. Kore (from Garki LGA)
274. Muku (from Garki LGA)
275. Raba (from Garki LGA)
276. Sabon Garin (from Garki LGA)
277. Wa (from Garki LGA)
278. Buduru (from Garki LGA)
279. Kafin Hausa (from Kafin Hausa LGA)
280. Kwazalewa (from Kafin Hausa LGA)
281. Sara (from Kafin Hausa LGA)
282. Azaga (from Kafin Hausa LGA)
283. Bulunchai (from Kafin Hausa LGA)
284. Daba (from Kafin Hausa LGA)
285. Daba Mashema (from Kafin Hausa LGA)
286. Fagi (from Kafin Hausa LGA)
287. Faragwai (from Kafin Hausa LGA)